

Relational data modeling

Module 5 - Lesson 1

Overview

Relational databases organize data into tables with relationships. Good modeling prevents duplicates, keeps data consistent, and makes queries simple and fast.

Key Concepts

- Tables have a primary key (usually an auto-incrementing id) that uniquely identifies rows.
- Foreign keys reference rows in other tables and enforce referential integrity.
- One-to-many: one course has many lessons; store the foreign key on the many side.
- Many-to-many needs a join table (e.g., students_courses) with two foreign keys.
- Normalize to remove redundancy, but denormalize selectively for performance.
- Index columns you filter or join on; indexes speed reads at the cost of slower writes.

Example

```
CREATE TABLE courses (  
  id SERIAL PRIMARY KEY,  
  title TEXT NOT NULL,  
  price NUMERIC(8,2)  
);  
CREATE TABLE lessons (  
  id SERIAL PRIMARY KEY,  
  course_id INTEGER REFERENCES courses(id),  
  title TEXT NOT NULL,  
  position INTEGER  
);
```

Summary

Relational modeling centers on primary keys, foreign keys, and well-defined relationships. A normalized schema with the right indexes keeps data consistent and queries fast.

Practice Checklist

- Model a course with modules, lessons, and students including a join table.
- Draw the relationships and identify which side holds each foreign key.
- Add an index to a column you filter on and explain the trade-off.